6	(a)		$B_{max} = \dfrac{\mu_0 I_{max}}{2\pi d}$ $= \dfrac{(4\pi \times 10^{-7})(1.2)}{(2\pi)(20 \times 10^{-2})}$ $= 1.2 \times 10^{-6}\ \text{T}\ (2\text{sf})$	C1 A1
	(b)	(i)	$\Phi_{max} = NB_{max}A = NB(\pi r^2)$ $= 1500(1.2 \times 10^{-6})\pi(0.50 \times 10^{-2})^2$ $= 1.41 \times 10^{-7}\ \text{Wb}\ (3\text{sf})$	C1 A1

		(ii)	The alternating current in the cable induces an alternating (changing) magnetic field at the plastic ring / toroidal solenoid. This cuts the solenoid and thus there is an alternating (changing) magnetic flux linkage at the solenoid. Hence, by Faraday's law of electromagnetic induction, e.m.f. is induced in the solenoid.	**B1** **B1**
		(iii)	The root-mean-square e.m.f. of the alternating voltage source is the equivalent to the value of steady direct current e.m.f. that dissipates the same power as the average amount of power dissipated by the alternating voltage. *(subtract 1m if average is not mentioned)*	**B1** **B1**

7	**(a)**		hc	**C1**